

Student's name

Student's number

Teacher's name



PLC PRESBYTERIAN
LADIES' COLLEGE
SYDNEY
1888

2015
TRIAL
HIGHER SCHOOL CERTIFICATE
EXAMINATION

Mathematics

General Instructions

- Reading time - 5 minutes
- Working time - 3 hours
- Write using blue or black pen
Black is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total Marks – 100

Section I: Pages 3-6

10 marks

- Attempt questions 1-10, using the answer sheet on page 23.
- Allow about 15 minutes for this section

Section II: Pages 7-19

90 marks

- Attempt questions 11-16, using the Answer Booklets provided.
- Allow about 2 hours 45 minutes for this section.

Multiple Choice	11	12	13	14	15	16	Total
							%

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Section I

10 marks

Attempt Questions 1 – 10.

Allow about 15 minutes for this section.

Use the multiple-choice answer sheet for Questions 1 – 10.

1. $\left(\frac{2a}{3b}\right)^{-5} = ?$

(A) $\frac{2a^5}{3b^5}$

(B) $\frac{3b^5}{2a^5}$

(C) $\frac{243b^5}{32a^5}$

(D) $\frac{1}{243b^5}$

2. Let α and β be the solutions of $2x^2 - 5x - 9 = 0$. Which value is the answer to $\frac{1}{\alpha} + \frac{1}{\beta}$?

(A) $-\frac{9}{2}$

(B) $-\frac{9}{5}$

(C) $-\frac{5}{9}$

(D) $\frac{5}{2}$

3. Which expression (value) is equal to $\lim_{x \rightarrow \infty} \frac{3x^2}{x^3 - x}$?
- (A) $\frac{3x}{x^2 - 1}$
- (B) 3
- (C) $\frac{3}{x - 1}$
- (D) 0
4. The period and amplitude of $y = 3 \cos 2x$ is:
- (A) Amplitude = 2 Period = $\frac{2\pi}{3}$
- (B) Amplitude = 3 Period = π
- (C) Amplitude = π Period = 3
- (D) Amplitude = $\frac{2\pi}{3}$ Period = 2
5. What is the value of x in the equation $\log_a 12 - 2\log_a 2 = \log_a x$?
- (A) 6
- (B) $\frac{1}{3}$
- (C) 3
- (D) $\frac{1}{6}$

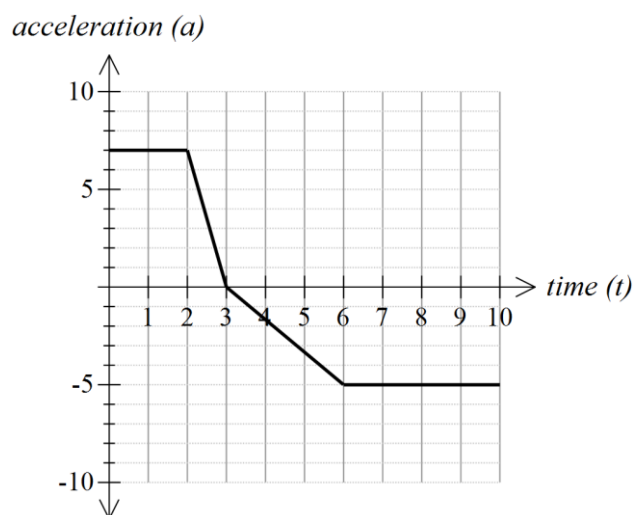
6. Which expression shows $\cos^2\left(\frac{\pi}{2} - \theta\right)\cot\theta$ simplified fully?
- (A) $\cos^2 \theta \cot \theta$
- (B) $\sin \theta \cos \theta$
- (C) $\frac{\sin^3 \theta}{\cos \theta}$
- (D) $\sin^2 \theta \cot \theta$
7. Which expression is equal to $\int_2^7 \frac{5}{x} dx$?
- (A) $5(\log_e 7 - \log_e 2)$
- (B) $\frac{1}{5}(\log_e 7 - \log_e 2)$
- (C) $\frac{5}{49} - \frac{5}{4}$
- (D) 0
8. Which expression is the equation of the normal to the curve $x^2 = 4y$ at the point where $x = 2$?
- (A) $y = 1$
- (B) $x - y - 1 = 0$
- (C) $y = -1$
- (D) $x + y - 3 = 0$

9. The function of $g(x)$ is given by

$$g(x) = \begin{cases} x^2 - 4 & \text{for } x > 0 \\ (X) & \text{for } (Y) \end{cases}$$

Which expressions for (X) and (Y) are correct, if $g(x)$ is an odd function?

- (A) $(X): 4 - x^2, (Y): x < 0$
- (B) $(X): -x^2 - 4, (Y): x < 0$
- (C) $(X): 4 - x^2, (Y): x > 0$
- (D) $(X): -x^2 - 4, (Y): x > 0$
10. A particle moves along a straight line. Initially it is at rest at the origin. The graph shows the acceleration, a , of the particle as a function of time t seconds for $0 \leq t \leq 10$.



At what time during the interval $0 \leq t \leq 10$ is the particle furthest from the origin?

- (A) 3 seconds
- (B) 6 seconds
- (C) 7 seconds
- (D) 8 seconds

End of Section I

Section II

90 marks

Attempt Questions 11 – 16.

Allow about 2 hours and 45 minutes for this section.

Answer each question in a **new writing booklet**. Extra writing booklets are available.

In Questions 11 – 16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a new writing booklet for Question 11.

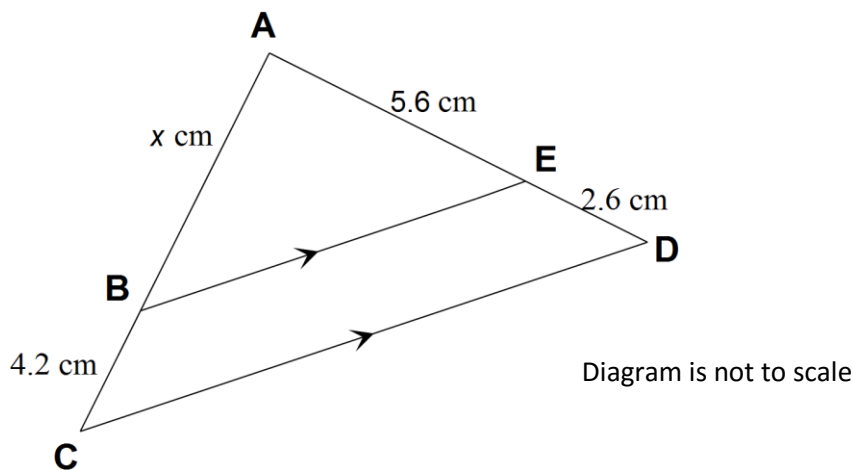
(a) Solve $x^2 - 2x - 7 = 0$, expressing your answer in simplest surd form. 2

(b) Find $\int \frac{3x}{x^2 + 1} dx$. 1

(c) Simplify fully : 2

$$\frac{2}{\sqrt{7} + 3} - \frac{3\sqrt{7}}{\sqrt{7} - 3}$$

(d) Find the value of x (correct to the nearest mm). 2



(e) Find the coordinates of the vertex and focus of the parabola $x^2 - 5y + 5 = 0$. 2

Question 11 continues on page 8

Question 11 (continued)

- (f) Water flows into an empty container, so that after t minutes the volume V of water in litres is given by **3**

$$V = \frac{12t^2}{t+4} \text{ for } t \geq 0.$$

What is the rate at which the water is flowing into the container at 1 minute?

- (g) Evaluate $\int_0^{\ln 6} e^x dx$. **2**

- (h) Differentiate $y = \sin 4x$ **1**

End of Question 11

Question 12 (15 marks) Use a new writing booklet for Question 12.

(a) Differentiate:

(i) $y = x^3 e^{3x}$. 2

(ii) $y = \frac{e^x}{(x+3)^2}$. (Full simplification of your answer is not required.) 2

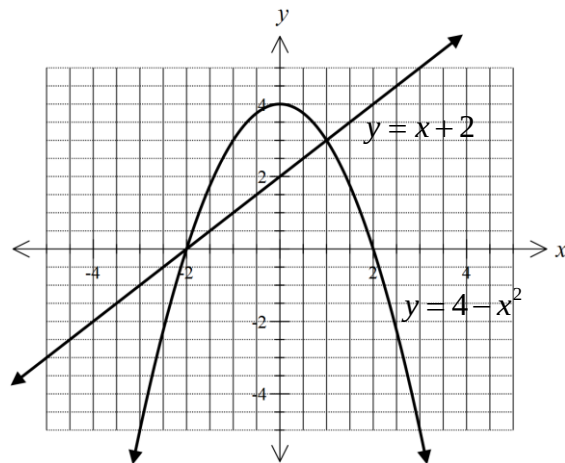
(b) Solve $\sqrt{3} \cos x = \sin x$ for $0 \leq x \leq 2\pi$. 2

(c) Use Simpson's Rule with five function values (x is in radians) to find an approximation for $\int_0^1 \tan x \, dx$. 2

(d) Evaluate $\int_0^{\frac{\pi}{2}} \sec^2 3x \, dx$ 2

Question 12 continues on page 10

- (e) Use the graphs below to answer (i) and (ii).



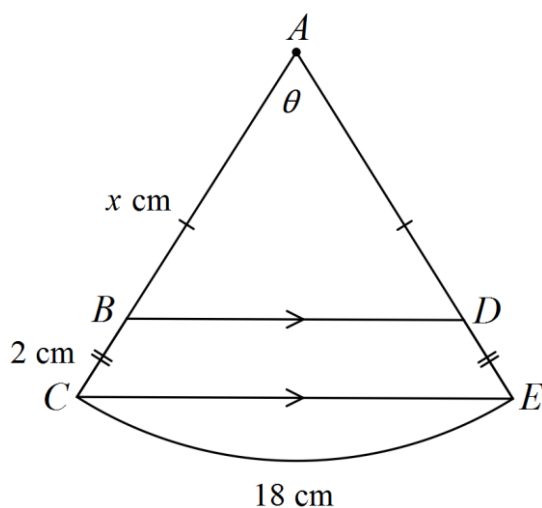
- (i) Solve the inequality $4 - x^2 \leq x + 2$. **1**
- (ii) Calculate the area between the curve $y = 4 - x^2$ and the line $y = x + 2$. **2**
- (f) Find the values of A , B and C if $3x^2 + x + 1 \equiv A(x - 1)(x + 2) + B(x + 1) + C$. **2**

End of Question 12

Question 13 (15 marks) Use a new writing booklet for Question 13.

(a) The diagram shows $\triangle ABD$ and $\triangle ACE$, where BD is parallel to CE ,

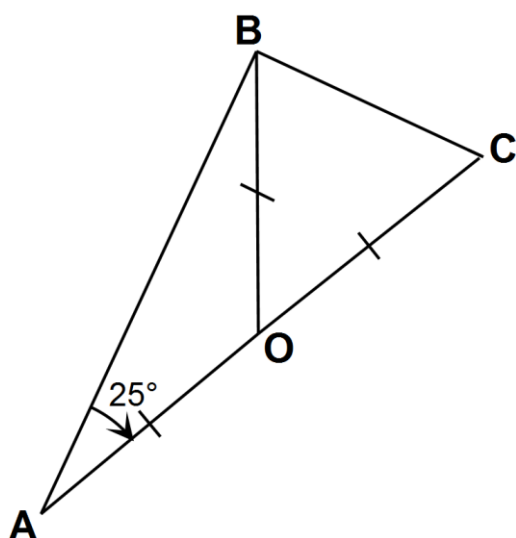
$AB = AD = x$ cm, $BC = DE = 2$ cm and $AD : AE = 3 : 4$. Triangle ACE and arc CE form a sector in a circle of radius $(x+2)$ cm. The angle of the sector is θ radians and arc $CE = 18$ cm.



(i) Find the value of θ . 2

(ii) Calculate the area of the segment cut off by CE . 2

(b) In the diagram below, $OA = OB = OC$. Show that $\angle OBC = 65^\circ$. Give reasons. 2

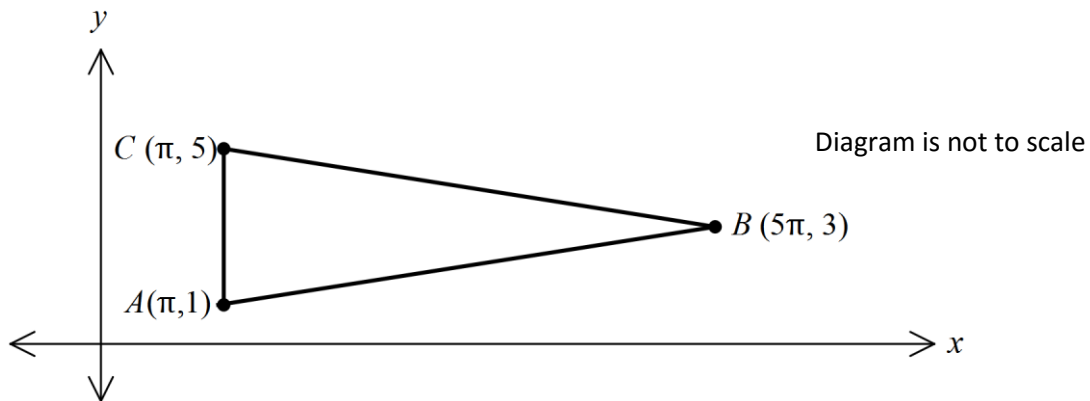


Question 13 continues on page 12

Question 13 (continued)

- (c) For the domain $0 \leq x \leq 6$, a function $y = f(x)$ satisfies $f'(x) < 0$ and $f''(x) < 0$. 2
Sketch a possible graph of $y = f(x)$ in this domain.

- (d) The points $A(\pi, 1)$, $B(5\pi, 3)$ and $C(\pi, 5)$ form an isosceles triangle, with $AB = BC$.

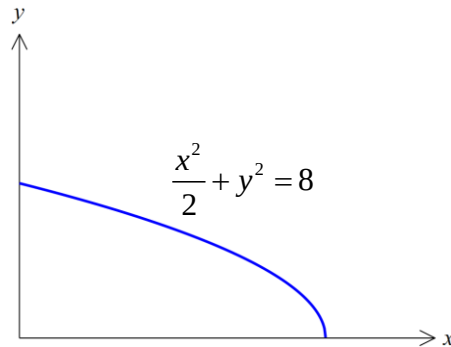


- (i) Find the midpoint of AB . 1
- (ii) Show that the equation of the line which is perpendicular to AB and which passes through point C is: 2
 $y + 2\pi x - 5 - 2\pi^2 = 0$
- (iii) Calculate the distance AB . 1
- (iv) Using the distances AB , BC and AC , or otherwise, find $\angle CAB$ to the nearest degree. 3

End of Question 13

Question 14 (15 marks) Use a new writing booklet for Question 14.

- (a) The part of the curve $\frac{x^2}{2} + y^2 = 8$ that lies in the first quadrant is drawn below. 2



This part of the curve is rotated about the **x-axis** to form a solid. Find the exact volume of this solid of revolution.

- (b) For the curve $y = x^3(3 - x)$
- (i) Find all stationary points and determine their nature. 3
 - (ii) Draw a sketch of the curve showing the stationary points, inflexion points and intercepts on the axes. 3

- (c) The displacement of a particle moving along the x -axis is given by

$$x = 5 \sin \frac{\pi}{2} t$$

where x is the displacement from the origin in metres, t is the time in minutes and $t \geq 0$.

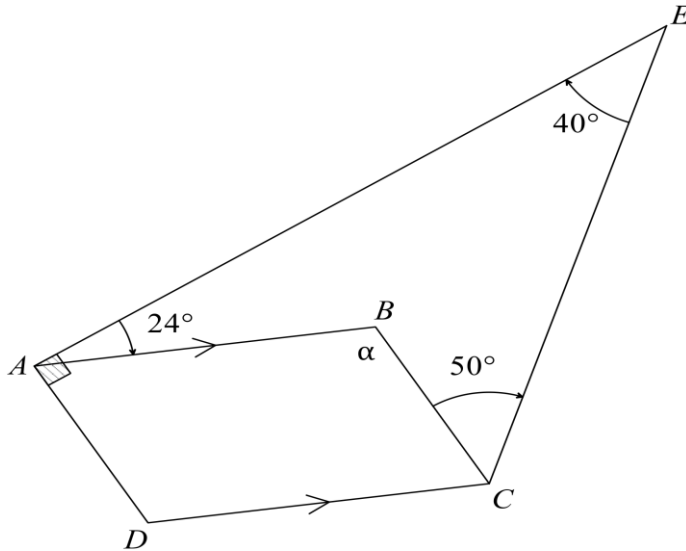
- (i) What is the furthest distance the particle moves away from the origin? 1
- (ii) When does the particle first return to its starting position? 1
- (iii) Find the acceleration of the particle when $t = 3 \text{ min}$. 2

Question 14 continues on page 14

Question 14 (continued)

- (d) In the quadrilateral $AECD$, $\angle DAE = 90^\circ$, $\angle AEC = 40^\circ$, $\angle BAE = 24^\circ$ and $\angle BCE = 50^\circ$.

In quadrilateral $ABCD$, AB is parallel to DC and $\angle ABC = \alpha$ as shown in the diagram.

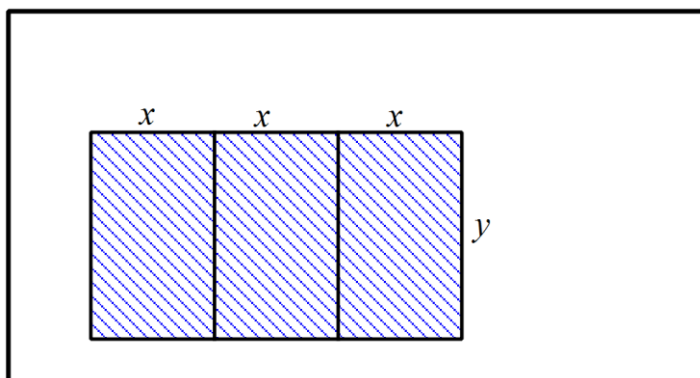


- | | | |
|------|---------------------------------------|----------|
| (i) | Explain why $\alpha = 114^\circ$. | 1 |
| (ii) | Prove that $ABCD$ is a parallelogram. | 2 |

End of Question 14

Question 15 (15 marks) Use a new writing booklet for Question 15.

- (a) Greg has a one hectare block of land ($10\,000\text{ m}^2 = 1\text{ hectare (ha)}$). He is going to fence off three identical rectangular plots within his block for his three children. Each plot will measure $x\text{ m}$ by $y\text{ m}$ as shown in the diagram below. He will retain the remainder of the block for himself and his wife. Greg can only afford 300 m of fencing to go around the children's plots.



- (i) Show that $y = 75 - \frac{3x}{2}$. **1**
- (ii) Find the value of x for which the area of the children's plots will be a maximum. **3**
- (iii) Find the maximum area of one of the children's blocks. **1**
- (iv) How much of Greg's 1 hectare block is left for him and his wife? **1**

Question 15 continues on page 16

Question 15 (continued)

- (b) The acceleration, after t seconds, of a particle moving in a straight line is given by

$$\ddot{x} = -\frac{14}{(t+4)^3}.$$

Initially the particle is located $\frac{3}{4}$ m to the left of the origin and the initial velocity is $\frac{7}{16}$ m/s.

- (i) Find the velocity v and the displacement x at any time t . 2
- (ii) What is the velocity of the particle when it passes through the origin? 2
- (iii) Sketch a graph of the displacement as a function of time. 2

- (c) A curve is given by the equation $y = 2x^{\frac{5}{2}} - x^3$, where $x \geq 0$.

- (i) Show that $\frac{d^2y}{dx^2} = \frac{15}{2}\sqrt{x} - 6x$. 1
- (ii) For what value(s) of x is the curve $y = 2x^{\frac{5}{2}} - x^3$ concave up? 2

End of Question 15

Question 16 (15 marks) Use a new writing booklet for Question 16.

- (a) Connor buys a new car, which begins to depreciate immediately. The value (\$ V) of the car after t years is given by $V = A e^{-kt}$

Where:

A is the initial value

k is the constant of depreciation

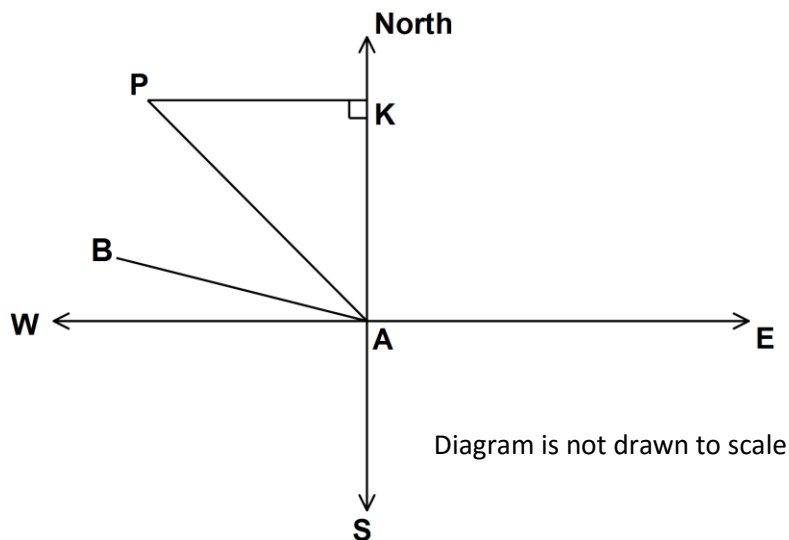
t is the time in years

The car is worth \$30 000 after 5 years and \$18 000 after 10 years.

- | | | |
|-------|---|----------|
| (i) | Find the constant of depreciation k . | 3 |
| (ii) | Find the initial value of the car. | 1 |
| (iii) | How many whole years will it take before the car's value falls below \$1 000? | 2 |

Question 16 continues on page 18

- (b) A plane leaves an airport (A) and travels due north $190\sqrt{3}$ kilometres to a point K and then turns due west and travels a further 190 kilometres until it reaches a point P . Due to storms the plane is then diverted to a new airport (B) which is 200 kilometres on a bearing of 280° from A .

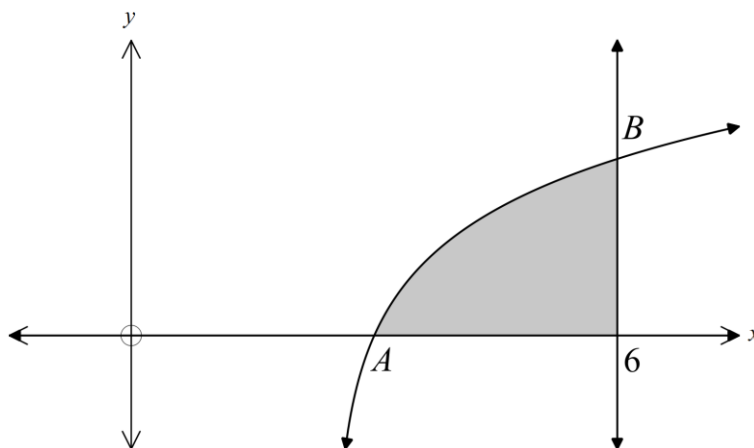


- | | | |
|-------|--|----------|
| (i) | Draw the diagram in your answer booklet and label it to show the information. | |
| (ii) | Show that $\angle KAP = 30^\circ$. | 1 |
| (iii) | Show that the plane needs to travel 294 kilometres from P to the new airport (B). | 2 |
| (iv) | Hence or otherwise find the bearing (to the nearest degree) on which the plane flies from P to B . | 1 |

Question 16 continues on page 19

- (c) The diagram shows a shaded region which is bounded by the curve $y = \log_e(2x - 5)$, the x axis and the line $x = 6$.

The curve $y = \log_e(2x - 5)$ intersects the x axis at A and the line $x = 6$ at B .



- (i) Show that the coordinates of points A and B are $(3, 0)$ and $(6, \log_e 7)$ respectively. **1**
- (ii) Show that if $y = \log_e(2x - 5)$, then $x = \frac{e^y + 5}{2}$. **1**
- (iii) Hence find the exact area of the shaded region. **3**

End of Examination

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STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

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Mathematics: Multiple Choice Answer Sheet

Student Number_____

Completely fill the response oval representing the most correct answer.

1. A ☐ B ☐ C ☐ D ☐

2. A ☐ B ☐ C ☐ D ☐

3. A ☐ B ☐ C ☐ D ☐

4. A ☐ B ☐ C ☐ D ☐

5. A ☐ B ☐ C ☐ D ☐

6. A ☐ B ☐ C ☐ D ☐

7. A ☐ B ☐ C ☐ D ☐

8. A ☐ B ☐ C ☐ D ☐

9. A ☐ B ☐ C ☐ D ☐

10. A ☐ B ☐ C ☐ D ☐

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Academic Year	12 Trial	Calendar Year	2015
Course	Unit 1 Maths	Name of task/exam	Trial HSC

Multiple choice

$$1. \left(\frac{2a}{3b}\right)^{-5} = \left(\frac{3b}{2a}\right)^5 = \frac{243b^5}{32a^5} \quad (C)$$

$$2. \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{5/2}{-9/2} = -\frac{5}{9} \quad (C)$$

$$3. \lim_{x \rightarrow \infty} \frac{3x^2}{x^3 - x} = \lim_{x \rightarrow \infty} \frac{\frac{3x^2}{x^3}}{\frac{x^3}{x^3} - \frac{x}{x^3}} = \lim_{x \rightarrow \infty} \frac{\frac{3}{x}}{1 - \frac{1}{x^2}} = 0 \quad (D)$$

4. (B)

5.

$$\begin{aligned} \log_a 12 &= \log_a 2^2 = \log_a \left(\frac{12}{4}\right) \\ &= \log_a 3 \\ &= \log_a x \\ \therefore \underline{x=3} \end{aligned}$$

(E)

$$6. \cos^2\left(\frac{\pi}{2} - \theta\right) \times \frac{\cos \theta}{\sin \theta} = \sin^2 \theta \times \frac{\cos \theta}{\sin \theta} = \sin \theta \cos \theta \quad (B)$$

$$\begin{aligned} 7. \int_2^7 \frac{5}{x} dx &= 5 \ln x \Big|_2^7 \\ &= 5 \ln 7 - 5 \ln 2 \end{aligned} \quad (A)$$

Solutions for exams and assessment tasks

Academic Year	12 Trial	Calendar Year	2015
Course	Unit Maths	Name of task/exam	Trial HSC

8. $y = \frac{x^2}{4}$

$$\frac{dy}{dx} = \frac{x}{2}$$

at $x=2$

$$m_T = 1 \quad m_N = -1$$

(D)

$$y-1 = -(x-2)$$

$$y-1 = -x+2$$

$$x+y-3=0$$

9. odd when $f(x) = -f(-x)$

$$g(x) = x^2 - 4$$

$$\begin{aligned} -g(-x) &= -(x^2 - 4) \\ &= -x^2 + 4 \end{aligned}$$

(A)

10. (D) Area under graphs above and below axes are equal.

Academic Year	12 Term 1	Calendar Year	2015
Course	Unit Maths	Name of task/exam	Final HSC

Question 11

a. $x^2 - 2x - 7 = 0$

$$x = \frac{2 \pm \sqrt{(-2)^2 - 4(1)(-7)}}{2}$$

$$= \frac{2 \pm \sqrt{32}}{2}$$

$$= \frac{2 \pm 4\sqrt{2}}{2}$$

$$= 1 \pm 2\sqrt{2}$$

b. $\int \frac{3x}{x^2+1} dx = \frac{3}{2} \int \frac{2x}{x^2+1} dx$

$$= \frac{3}{2} \ln(x^2+1) + C$$

c. $\frac{2(\sqrt{7}-3) - 3\sqrt{7}(\sqrt{7}+3)}{7-9} = \frac{2\sqrt{7}-6-21-9\sqrt{7}}{-2}$

$$= \frac{-7\sqrt{7}-27}{-2}$$

$$= \frac{7\sqrt{7}+27}{2}$$

d. $\frac{x}{4.2} = \frac{5.6}{2.6}$

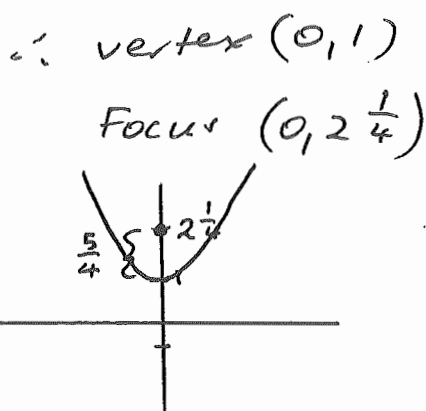
$$x = \frac{5.6}{2.6} \times 4.2$$

$$x = 9.04 \text{ cm}$$

$$= 90 \text{ mm}$$

Academic Year	T2 Trial	Calendar Year	2015
Course	Unit 1 Math	Name of task/exam	Trial HSC

e. $x^2 - 5y + 5 = 0$
 $5y = x^2 + 5$
 $x^2 = 5y - 5$
 $x^2 = 5(y - 1)$
 $[x^2 = 4a(y - k)]$
 $5 = 4a$
 $a = \frac{5}{4}$



f. $V = \frac{12t^2}{t+4} \quad t \geq 0$

$u = 12t^2 \quad v = t+4$
 $u' = 24t \quad v' = 1$

$$\frac{dV}{dt} = \frac{24t(t+4) - 12t^2}{(t+4)^2}$$

$$= \frac{24t^2 + 96t - 12t^2}{(t+4)^2}$$

at $t = 1$

$$\frac{dV}{dt} = \frac{24 + 96 - 12}{5^2}$$

$$= \frac{108}{25} = 4.32 \text{ L/min}$$

g. $\int_0^{\ln 6} e^x dx = e^x \Big|_0^{\ln 6}$
 $= 6 - 1$
 $= 5$

h. $\frac{dy}{dx} = 4 \cos 4x$

Academic Year	12 Trial	Calendar Year	2015
Course	Zenith Maths	Name of task/exam	Trial HSC

Question 12

a (i) $y = x^3 e^{3x}$

$$u = x^3 \quad v = e^{3x}$$

$$u' = 3x^2 \quad v' = 3e^{3x}$$

$$\frac{dy}{dx} = 3x^2 e^{3x} + 3x^3 e^{3x}$$

$$= 3x^2 e^{3x} (1 + x)$$

(ii) $y = \frac{e^x}{(x+3)^2}$

$$u = e^x \quad v = (x+3)^2$$

$$u' = e^x \quad v' = 2(x+3)$$

$$= 2x+6$$

$$\frac{dy}{dx} = \frac{e^x (x+3)^2 - 2e^x (x+3)}{(x+3)^4}$$

$$= \frac{e^x (x+3) (x+3-2)}{(x+3)^4}$$

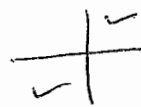
$$= \frac{e^x (x+3) (x+1)}{(x+3)^4}$$

$$= \frac{e^x (x+1)}{(x+3)^3}$$

b. $\sqrt{3} \cos x = \sin x \quad (\div \cos x)$

$$\tan x = \sqrt{3}$$

$$x = \frac{\pi}{3}, \frac{4\pi}{3}$$



c.

0	$\frac{1}{4}$	$\frac{2}{4}$	$\frac{3}{4}$	1
0	$\tan \frac{1}{4}$	$\tan \frac{2}{4}$	$\tan \frac{3}{4}$	$\tan 1$
y_0	y_1	y_2	y_3	y_n

$$A = \frac{1}{3} \left[0 + \tan 1 + 4(\tan \frac{1}{4} + \tan \frac{3}{4}) + 2(\tan \frac{1}{2}) \right]$$

$$= \frac{1}{12} [7.3977 \dots]$$

$$= 0.6164805 \dots$$

$$\approx 0.62 \quad (2dp)$$

Academic Year	12 Trial	Calendar Year	2015
Course	Unit Maths	Name of task/exam	Trial HSC

$$\begin{aligned}
 d. \int_0^{\frac{\pi}{2}} \sec^2 3x \, dx &= \frac{1}{3} \tan 3x \Big|_0^{\frac{\pi}{2}} \\
 &= \frac{1}{3} \tan \frac{3\pi}{2} - \frac{1}{3} \tan 0 \\
 &= \frac{1}{3} \tan \frac{3\pi}{2} \\
 &= \text{undefined.}
 \end{aligned}$$

$$e. (i) x \leq -2, x \geq 1$$

$$\begin{aligned}
 (ii) \int_{-2}^1 4 - x^2 - (x+2) \, dx &= \int_{-2}^1 4 - x^2 - x - 2 \, dx \\
 &= \int_{-2}^1 2 - x - x^2 \, dx \\
 &= \left[2x - \frac{x^2}{2} - \frac{x^3}{3} \right]_{-2}^1 \\
 &= \left(2(1) - \frac{1}{2} - \frac{1}{3} \right) - \left(2(-2) - \frac{4}{2} - \frac{(-2)^3}{3} \right) \\
 &= \frac{7}{6} - -\frac{10}{3} \\
 &= \frac{9}{2} \text{ units}^2
 \end{aligned}$$

$$\begin{aligned}
 f. \quad 3x^2 + x + 1 &\equiv A(x^2 + x - 2) + Bx + C + C \\
 &\equiv Ax^2 + (A+B)x - 2A + B + C
 \end{aligned}$$

$$\therefore \quad A = 3 \quad \begin{matrix} A+B=1 \\ \textcircled{1} \end{matrix} \quad \begin{matrix} -2A+B+C=1 \\ \textcircled{2} \end{matrix}$$

$$\begin{aligned}
 \textcircled{1} \quad A+B &= 1 \\
 3+B &= 1 \\
 \underline{B} &= -2
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{2} \quad -2A+B+C &= 1 \\
 -2(3)-2+C &= 1 \\
 -8+C &= 1 \\
 \underline{C} &= 9
 \end{aligned}$$

$$\therefore \quad A = 3 \quad B = -2 \quad C = 9.$$

Academic Year	12 Trial	Calendar Year	2015
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Question 13

$$(i) \quad (1) \quad l = r\theta$$

$$18 = r \times \theta$$

$$\text{Since } x = 6$$

$$r = 8$$

$$\therefore 18 = 8 \times \theta$$

$$\theta = \frac{18}{8} = \frac{9}{4}$$

$$(2) \quad \frac{AD}{AE} = \frac{3}{4}$$

$$\frac{x}{x+2} = \frac{3}{4}$$

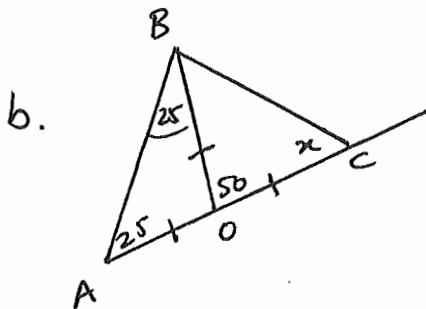
$$4x = 3(x+2)$$

$$x = 6 \text{ cm.}$$

$$(ii) \quad A = \frac{1}{2} r^2 (\theta - \sin \theta)$$

$$= \frac{1}{2} \times 8^2 \left(\frac{9}{4} - \sin \frac{9}{4} \right)$$

$$= 8 \left(\frac{9}{4} - \sin \frac{9}{4} \right) \doteq 47.1 \text{ cm}^2$$



$$OA = OB = OC \quad (\text{equal given})$$

$\triangle AOB$ is an isosceles $\triangle$.

$$\therefore \angle OAB = \angle OBA = 25^\circ \quad (\text{base angles equal})$$

$$\angle BOC = 50^\circ \quad (\text{exterior angle of } \triangle)$$

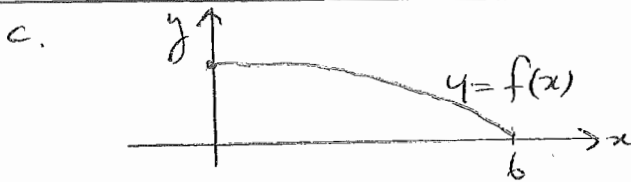
$$\angle OBC = \frac{180 - 50}{2} \quad (\text{angle sum of } \triangle)$$

(base angles equal in isosceles $\triangle$)

$$\angle OBC = \frac{130}{2}$$

$$= 65^\circ$$

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$$\begin{aligned} \text{d. (i) midpoint } AB &= \left(\frac{\pi + 5\pi}{2}, \frac{1+3}{2} \right) \\ &= (3\pi, 2) \end{aligned}$$

$$A(\pi, 1)$$

$$B(5\pi, 3)$$

$$C(\pi, 5)$$

$$(ii) m_{AB} = \frac{3-1}{5\pi-\pi} = \frac{2}{4\pi} = \frac{1}{2\pi}$$

$$\perp \text{ to } AB: m_{\perp} = -2\pi$$

$$m = -2\pi \quad C(\pi, 5)$$

$$y - 5 = -2\pi(x - \pi)$$

$$y = -2\pi x + 2\pi^2 + 5$$

$$y + 2\pi x - 5 - 2\pi^2 = 0$$

$$\begin{aligned} (iii) d_{AB} &= \sqrt{(5\pi - \pi)^2 + (3 - 1)^2} \\ &= \sqrt{16\pi^2 + 4} \\ &= \sqrt{4(4\pi^2 + 1)} \\ &= 2\sqrt{4\pi^2 + 1} \end{aligned}$$

(iv) $\triangle ABC$ is an isosceles $\triangle$

$$\therefore AB = BC = 2\sqrt{4\pi^2 + 1}, \quad AC = 4$$

$$\begin{aligned} \cos \angle CAB &= \frac{4^2 + (2\sqrt{4\pi^2 + 1})^2 - (2\sqrt{4\pi^2 + 1})^2}{2 \times 4 \times 2\sqrt{4\pi^2 + 1}} \\ &= \frac{16 + 4(4\pi^2 + 1) - 4(4\pi^2 + 1)}{16\sqrt{4\pi^2 + 1}} \\ &= \frac{16}{16\sqrt{4\pi^2 + 1}} \\ \angle CAB &= \cos^{-1} \left(\frac{1}{\sqrt{4\pi^2 + 1}} \right) \\ &\approx 81^\circ \quad (\text{nearest degree}) \end{aligned}$$

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Course	Unit Maths	Name of task/exam	Unit HSC

Question 14

a. $\frac{x^2}{2} + y^2 = 8$

$$y^2 = 8 - \frac{x^2}{2}$$

at $y = 0$.

$$0 = 8 - \frac{x^2}{2}$$

$$\frac{x^2}{2} = 8$$

$$x = \pm 4.$$

$$\begin{aligned} \therefore V &= \pi \int y^2 dx \\ &= \pi \int_0^4 8 - \frac{x^2}{2} dx \end{aligned}$$

$$= \pi \left[8x - \frac{x^3}{6} \right]_0^4$$

$$= \pi \left[\left(32 - \frac{64}{6} \right) - (0) \right]$$

$$= \frac{64\pi}{3} \text{ units}^3$$

b. $y = x^3(3-x) = 3x^3 - x^4$

(i) $\frac{dy}{dx} = 9x^2 - 4x^3$

$$\frac{d^2y}{dx^2} = 18x - 12x^2$$

$$\frac{dy}{dx} = 9x^2 - 4x^3 = 0 \quad \text{for stat points}$$

$$= x^2(9 - 4x)$$

$$= 0$$

$$\begin{aligned} \therefore x &= 0 & 9 - 4x &= 0 \\ & & x &= \frac{9}{4} \end{aligned}$$

Academic Year	12	Calendar Year	2015
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at $x=0$ $\frac{d^2y}{dx^2} = 18(0) - 72(0)$
 $\frac{d^2y}{dx^2} = 0$

$\therefore$ possible point of inflexion at $x=0$.

at $x = \frac{9}{4}$ $\frac{d^2y}{dx^2} = 18\left(\frac{9}{4}\right) - 72\left(\frac{9}{4}\right)^2$
 $= -\frac{81}{4} < 0 \therefore \text{max.}$

at $x=0$:

$y'' = \begin{vmatrix} -1 & 0 & 0 \\ -30 & 0 & 6 \end{vmatrix}$

$\therefore$ point of inflexion (concavity change)

$\therefore$ horizontal point of inflexion

Since $\frac{dy}{dx} = 0$ AND $\frac{d^2y}{dx^2} = 0$

at $x=0$:

$y=0$

$(0,0)$ inflex
horiz

at $x = \frac{9}{4}$: $y = \left(\frac{9}{4}\right)^3 \left(3 - \frac{9}{4}\right)$

$= \frac{2187}{256} \doteq 8.543$

$\left(\frac{9}{4}, 8.543\right)$ MAX.

(ii) at $x=0$ $y=0$ $(0,0)$.

inflexions: $18x - 12x^2 = 0$

$6x(3 - 2x) = 0$

$x=0$ $x = \frac{3}{2}$

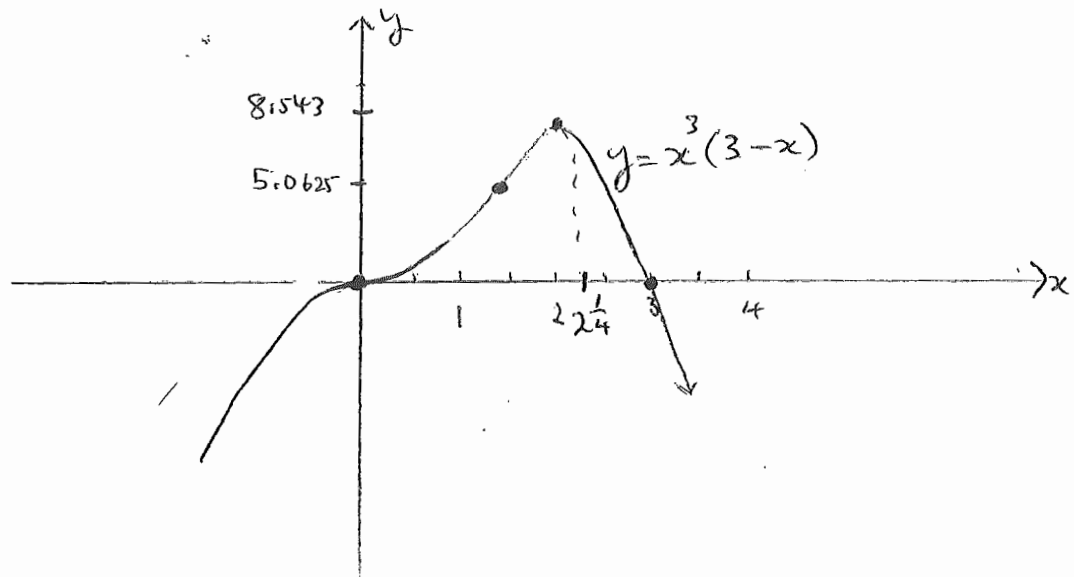
Academic Year	12	Calendar Year	2015
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at $x = \frac{3}{2}$

x	1	1.5	2
y''	6	0	-12
	↑		↓

$\therefore$ change in concavity

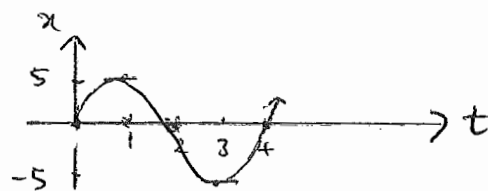
$\therefore$ point of inflexion at $(\frac{3}{2}, 5.0625)$



at $y=0$ $x^3(3-x)=0$
 $\therefore x=0$ $x=3$ $(0,0)$ $(3,0)$

c. $x = 5 \sin \frac{\pi}{2} t$

(i) 5 metres



(ii) From graph: after 2 minutes

OR $x=0 \therefore 5 \sin \frac{\pi}{2} t = 0$

$$\sin \frac{\pi}{2} t = 0$$

$$\frac{\pi}{2} t = 0, \pi, 2\pi, \dots$$

$$t = 0, 2, 4, \dots$$

$\therefore$ after 2 minutes...

Solutions for exams and assessment tasks

Academic Year	12	Calendar Year	2015
Course	Unit + Trial	Name of task/exam	Trial HSC

$$(iii) \quad x = 5 \sin \frac{\pi}{2} t$$

$$\dot{x} = \frac{5\pi}{2} \cos \frac{\pi}{2} t$$

$$\ddot{x} = -5\left(\frac{\pi}{2}\right)^2 \sin \frac{\pi}{2} t$$

$$\text{at } t = 3 \quad \ddot{x} = -5\left(\frac{\pi}{2}\right)^2 \sin \frac{3\pi}{2}$$

$$= -5 \frac{\pi^2}{4} \times -1$$

$$= \frac{5\pi^2}{4}$$

$$d. (i) \text{ reflex } \angle ABC = 360 - (24 + 50 + 40) = 246^\circ$$

$$\alpha = 360 - 246 = 114^\circ$$

$$(ii) \text{ Since } \alpha = 114^\circ$$

$$\begin{aligned} \angle BCD &= 180 - 114 \\ &= 66^\circ \end{aligned}$$

(co-interior angles are supplementary on parallel lines)

$$\begin{aligned} \therefore \angle ECD &= 66 + 50 \\ &= 116^\circ \end{aligned}$$

$$\begin{aligned} \angle ADC &= 360 - 116 - 90 - 40 \quad (\text{angle sum of quadrilateral}) \\ &= 114^\circ \end{aligned}$$

$$\begin{aligned} \angle BAD &= 90 - 24^\circ \quad (\text{given}) \\ &= 66^\circ \end{aligned}$$

$$\begin{aligned} \angle ADC + \angle BCD &= 66 + 114 \\ &= 180^\circ \end{aligned}$$

$$\therefore BC \parallel AD \quad (\text{co-interior angles exist})$$

$\therefore$ Since opposite angles are equal and opposite sides are parallel

Then ABCD is a parallelogram.

Academic Year	12	Calendar Year	2015
Course	Unit Math	Name of task/exam	Trial HSC

Question 15

$$(i) \quad 6x + 4y = 300$$

$$4y = 300 - 6x$$

$$y = \frac{300 - 6x}{4}$$

$$y = \frac{300}{4} - \frac{6x}{4}$$

$$y = 75 - \frac{3x}{2}$$

$$(ii) \quad A = 3xy$$

$$A = 3x \left(75 - \frac{3x}{2} \right)$$

$$\frac{dA}{dx} = 3x \left(-\frac{3}{2} \right) + \left(75 - \frac{3x}{2} \right) \times 3$$

$$= -\frac{9x}{2} + 225 - \frac{9x}{2}$$

$$= -\frac{18x}{2} + 225$$

$$\frac{d^2A}{dx^2} = -\frac{18}{2} < 0 \quad \therefore \text{max.}$$

$$\frac{dA}{dx} = 0 \quad \text{for stat. pt.}$$

$$-\frac{18x}{2} + 225 = 0$$

$$x = \frac{225}{9}$$

$$\therefore \text{ at } x = \frac{225}{9} \text{ maximum area}$$

$$x = 25$$

$$u = 3x \quad v = 75 - \frac{3x}{2}$$

$$u' = 3 \quad v' = -\frac{3}{2}$$

Academic Year	12	Calendar Year	2015
Course	Unit Maths	Name of task/exam	Trial HSC

$\therefore$ maximum area of one of the children's

blocks : $A = xy$

$$\text{at } x = 25, \quad y = 75 - \frac{3}{2}(25)$$

$$= \frac{75}{2}$$

$$\therefore \text{Area} = 25 \times \frac{75}{2}$$

$$= 937.5 \text{ m}^2$$

$$\begin{aligned} \text{(iv) Area remaining} &= 10000 - 3 \times 937.5 \\ &= 7187.5 \text{ m}^2. \end{aligned}$$

Academic Year	12	Calendar Year	2015
Course	Unit Math	Name of task/exam	Trial HSC

Question 15 continued

$$b. \quad \ddot{x} = -\frac{14}{(t+4)^3}$$

$$\text{at } t=0 \quad x = -\frac{3}{4} \quad \dot{x} = \frac{7}{16} \text{ m/s.}$$

$$\begin{aligned} (i) \quad \dot{x} &= -14 \int \frac{1}{(t+4)^3} dt \\ &= -14 \int (t+4)^{-3} dt \end{aligned}$$

$$= -14 \left[\frac{(t+4)^{-2}}{-2} \right] + C$$

$$= 7(t+4)^{-2} + C$$

$$\text{at } t=0 \quad \dot{x} = \frac{7}{16}$$

$$\frac{7}{16} = 7(0+4)^{-2} + C$$

$$\frac{7}{16} = \frac{7}{16} + C$$

$$\therefore C = 0$$

$$\dot{x} = \frac{7}{(t+4)^2}$$

$$x = 7 \int (t+4)^{-2} dt$$

$$x = 7 \frac{(t+4)^{-1}}{-1} + C$$

$$\text{at } t=0 \quad x = -\frac{3}{4}$$

Academic Year	12	Calendar Year	2015
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$$-\frac{3}{4} = -7(4)^{-1} + C$$

$$C = -\frac{3}{4} + \frac{7}{4}$$

$$C = 1$$

$$\therefore x = -\frac{7}{t+4} + 1$$

(ii) at $x = 0$

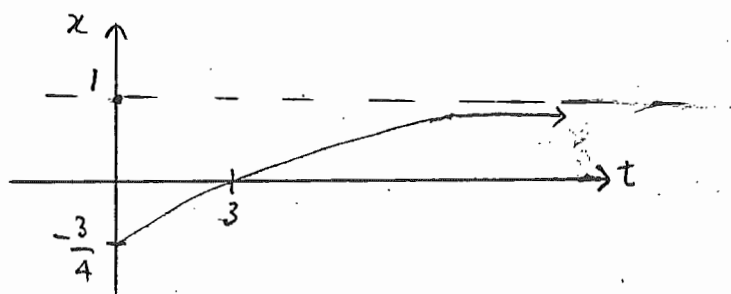
$$0 = -\frac{7}{t+4} + 1$$

$$0 = -7 + t + 4$$

$$\underline{t = 3}$$

$$\begin{aligned} \therefore x &= \frac{7}{(t+4)^2} \\ &= \frac{7}{(7)^2} \\ &= \frac{1}{7} \end{aligned}$$

(iii)



$$x = -\frac{7}{t+4} + 1$$

at $t = 0$ $x = -\frac{3}{4}$

Academic Year	12	Calendar Year	2015
Course	Unit Maths	Name of task/exam	Total HSC

c. $y = 2x^{\frac{5}{2}} - x^3$

(i) $\frac{dy}{dx} = 5x^{\frac{3}{2}} - 3x^2$

$$\frac{d^2y}{dx^2} = \frac{15}{2}x^{\frac{1}{2}} - 6x$$

$$= \frac{15\sqrt{x}}{2} - 6x$$

(ii) $\frac{d^2y}{dx^2} > 0$

$$\frac{15}{2}x^{\frac{1}{2}} - 6x > 0$$

$$3x^{\frac{1}{2}}\left(\frac{5}{2} - 2\sqrt{x}\right) > 0$$

$$\begin{array}{c} | \quad | \quad | \\ 0 \quad \text{(check)} \quad \frac{25}{16} \end{array}$$

$$\therefore 0 < x < \frac{25}{16}$$

$$-2\sqrt{x} + \frac{5}{2} = 0$$

$$2\sqrt{x} = \frac{5}{2}$$

$$4x = \frac{25}{4}$$

$$x = \frac{25}{16}$$

method 2:

$$\frac{15\sqrt{x}}{2} - 6x > 0$$

$$\frac{15\sqrt{x}}{2} > 6x$$

since $\sqrt{x} > 0$ and $x > 0$ (length)

then $\frac{225x}{4} > 36x^2$ (square both sides)

$$225x > 144x^2$$

$$144x^2 - 225x < 0$$

$$x(144x - 225) < 0$$

$$144x = 225$$

$$x = \frac{25}{16}$$

$$\begin{array}{c} | \quad | \\ 0 \quad \frac{25}{16} \end{array} \quad 0 \leq x \leq \frac{25}{16}$$

Academic Year	12	Calendar Year	2015
Course	2 Unit Maths	Name of task/exam	Trial FSL

Question 16.

a. $V = A e^{-kt}$

$$30000 = A e^{-5k}$$

$$18000 = A e^{-10k}$$

$$(i) \frac{A e^{-5k}}{A e^{-10k}} = \frac{30000}{18000}$$

$$e^{5k} = \frac{5}{3}$$

$$5k = \ln \frac{5}{3}$$

$$k = \frac{1}{5} \ln \left(\frac{5}{3} \right)$$

$$\approx 0.102165 \dots$$

(ii)

$$30000 = A e^{-5 \left(\frac{1}{5} \ln \left(\frac{5}{3} \right) \right)}$$

$$30000 = A e^{-\ln \left(\frac{5}{3} \right)}$$

$$A = \frac{30000}{\frac{3}{5}}$$

$$A = \$50000$$

note: $e^{-\ln \left(\frac{5}{3} \right)} = e^{\ln \frac{3}{5}} = \frac{3}{5}$

(iii)

$$1000 = 50000 e^{-k \times t}$$

$$\frac{1}{50} = e^{-k \times t}$$

$$-kt = \ln \left(\frac{1}{50} \right)$$

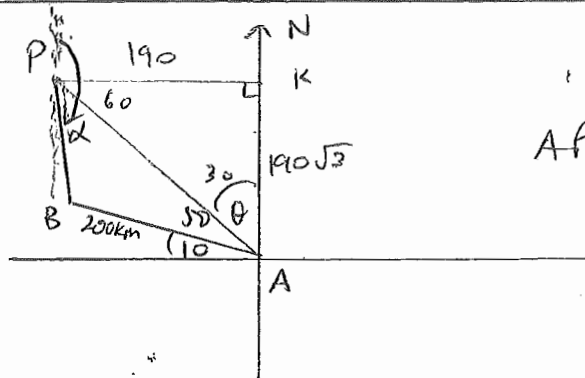
$$t = \frac{\ln \frac{1}{50}}{-\frac{1}{5} \ln \left(\frac{5}{3} \right)} \quad \text{where } k = \frac{1}{5} \left(\ln \left(\frac{5}{3} \right) \right)$$

$$t = 38.29 \dots$$

$\therefore$ 39 years for it to fall below \$1000

Academic Year	12	Calendar Year	2015
Course	Unit Maths	Name of task/exam	Trig HSC

b(i)



$$AP = \sqrt{190^2 + (190\sqrt{3})^2} = 380 \text{ km.}$$

$$(ii) \tan \theta = \frac{190}{190\sqrt{3}}$$

$$\tan \theta = \frac{1}{\sqrt{3}}$$

$$\theta = 30^\circ$$

$$\therefore \angle KAP = 30^\circ$$

$$(iii) \angle PAB = 50^\circ$$

$$PB^2 = 200^2 + 380^2 - 2 \times 200 \times 380 \times \cos 50^\circ$$

$$= 86696.283 \dots$$

$$PB = \sqrt{86696.283 \dots}$$

$$= 294.442326 \dots$$

$$\approx 294 \text{ km}$$

$$(iv) \angle APB : \frac{\sin \alpha}{200} = \frac{\sin 50^\circ}{294.442326 \dots}$$

$$\sin \alpha = \frac{200 \sin 50^\circ}{294.442326 \dots}$$

$$\alpha \approx 31^\circ 21'$$

$$\therefore \text{Bearing is } 180 + 31^\circ 21'$$

$$= \underline{181^\circ 21'}$$

Academic Year	12	Calendar Year	2015
Course	2 unit maths	Name of task/exam	Trinal HSC

c. (i) $y = \log_e(2x-5)$

at $x=6$ $y = \log_e(12-5)$

$y = \log_e 7$ $B(6, \ln 7)$

at $y=0$ $0 = \log_e(2x-5)$

$e^0 = 2x-5$

$1 = 2x-5$

$2x = 6$

$x = 3$

$A(3, 0)$

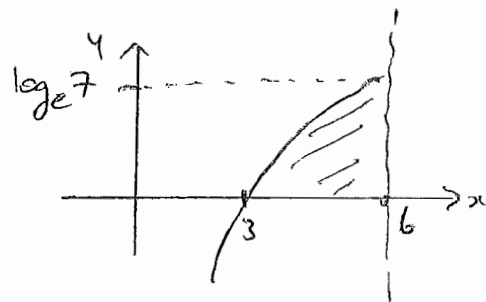
(ii) $y = \log_e(2x-5)$

$e^y = 2x-5$

$2x = e^y + 5$

$x = \frac{e^y + 5}{2}$

(iii) $\text{Area}_{\text{rect}} = 6 \times \log_e 7$
 $= 6 \log_e 7$



Area bound by y -axis:

$$\begin{aligned}
 A &= \int_0^{\ln 7} \frac{e^y + 5}{2} dy = \frac{1}{2} \left[(e^y + 5y) \right]_0^{\ln 7} \\
 &= \frac{1}{2} \left[e^{\ln 7} + 5 \ln 7 - (e^0 + 5(0)) \right] \\
 &= \frac{1}{2} [7 + 5 \ln 7 - 1] \\
 &= \frac{1}{2} [6 + 5 \ln 7] \\
 &= 3 + \frac{5}{2} \ln 7.
 \end{aligned}$$

$$\begin{aligned}
 \therefore \text{Shaded area} &= 6 \ln 7 - 3 - \frac{5}{2} \ln 7 \\
 &= \frac{7}{2} \ln 7 - 3.
 \end{aligned}$$